State-of-the-art Algorithms for Breast Cancer Detection: A Comparative Analysis

Trisha Batta
Department of Computer Science & Engineering
Chitkara University
Punjab, India
trisha2468.be22@chitkara.edu.in

Abstract— Breast cancer is a major health issue of concern to the population, as it is one of the most commonly diagnosed cancer in women across the globe. Tumor pathology identification is a critical aspect to consider when it comes to patient prognosis and treatment outcomes and must be identified as soon as possible and with the greatest accuracy. The study will examine and compare the diagnostic potential of three well-known supervised classification algorithms, namely, Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM) when used in binary tumor classification with the use of the Wisconsin Breast Cancer Diagnostic (WBCD) dataset. An experimental framework is used which is rigorous and standardized, including uniform data preprocessing, train-test splitting, and multi-metric performance evaluation using accuracy, precision, recall, and F1-score. The findings show that LR achieves the best accuracy of 98.60% and F1-score of 98.89%. SVM achieves 97.90%, CNN achieves 97.20%, and RF achieves the highest recall of 97.78%. A direct comparison with Naji et al. [13], who has applied the same three classical algorithms on WBCD, shows that the proposed framework yields improved results through uniform preprocessing and extended multi-metric evaluation. Guidelines on a practical implementation of these models in clinical decision-support systems are suggested according to the empirical results.

Keywords—breast cancer diagnosis; supervised classification; logistic regression; random forest; support vector machine; WBCD dataset; clinical decision support; machine learning in healthcare

I. Introduction

At the global level, breast cancer has remained as the most common form of cancer diagnosis in women and poses a significant burden to healthcare systems in both the industrialized and resource-limited regions. According to the WHO, it is estimated that there were 2.3 million new cases that were confirmed in 2022; and it resulted in more than 685,000 deaths [1]. The stage of the stage of identifying the condition is closely related to the prognosis, as patients with localized disease have a survival probability of more than 99 years, and with the metastatic disease, the survival rate is in the range of 30 at best.

The current gold-standard methods use mammographic imaging together with breast ultrasonography and fine needle aspiration (FNA) cytology. The three methods have demonstrated their value yet each method presents specific limitations. Radiological interpretation is very much dependent on the ability of the specialist performing the exam, which creates inter-practitioner variability and allows over-diagnosis as well as missed diagnostics. The infrastructural expenses of imaging equipment and qualified personnel are still prohibitive in most health care institutions especially in the low- and middle-income nations. Additional issues in creating diagnostic consistency in FNA cytology slides subjective visual analysis of the slides is what makes objective and reproducible classification tools highly desirable [2].

Machine learning (ML) can be viewed as a promising alternative avenue to more conventional diagnosis using it to identify complex discriminative patterns in high-dimensional clinical data. In comparison to rule-based expert systems, ML-based classifiers make predictions based on historic training examples and can make consistent and repeatable predictions, without human intervention in the inference time. The particular advantage of supervised classification models has been demonstrated in the context of medical applications, where there are labeled training examples, and the output is needed to be binary or multi-class restricting. Simultaneously considering dozens of cellular features simultaneously, the capacity of ML models can be useful in radiologists and pathologists, potentially assisting them with quantitative data-based second opinions [3].

Logistic Regression (LR), Random Forest (RF), and Support Vector Machine (SVM), and Convolutional Neural Networks (CNN) are in the list of popular supervised learners with a wide range of types of supervised learners: LR is more transparent and interpretable due to its use in regulated clinical settings; RF is more robust as it employs the ensemble-based approach and inherently ranks the features; SVM is more robust, as it is based on strong margin-based generalization, i.e., kernel transformations; CNN implemented here as a Multi-Layer Perceptron(MLP). Although there is a large amount of literature that studies individual algorithms, there is a relative dearth of rigorously controlled head-to-head comparisons on the WBCD benchmark, with uniform preprocessing conditions, of individual algorithms. These comparisons are necessary because practitioners should explain how they have chosen algorithms in clinical implementation.

The current research meets this requirement by providing a systematic, reproducible experiment on the application of LR, RF, SVM and CNN to WBCD data in

standard conditions. It majorly contributes by three factors: (i) a endogenously controlled multi-metric comparative assessment of three popular classifiers; (ii) a study of the influence of feature normalization on the performance of linear and non-linear model architecture; and (iii) evidence-based guidelines on algorithm choice in particular clinical deployment settings.

The rest of this paper is divided into the following sections; Section II is a literature review of the existing research on ML based breast cancer classification; Section III outlines data characteristics, preprocessing procedures, and classification strategies; Section IV presents the results of the experiment; Section V offers a comprehensive analysis and discussion of the implications; Section VI will conclude the paper and define the further research directions.

II. Related Work

Computational techniques in breast cancer detection Computational techniques have been a subject of study since the early 1980s, and at least since that time the WBCD dataset has been used as a reference point since its original publication by Wolberg and Mangasarian [2]. Their ground breaking advance focused on the use of linear programming methods to isolate benign and malignant cell nuclei features with an accuracy of over 97 in classification. Later Bennett and Mangasarian [3] elaborated this structure identically with the tools of robust optimization, setting strong baselines to be the inspiration of decades of subsequent research.

Some of the earliest approaches that were examined to solve this problem include neural network based approaches. Setiono [11] showed that aggressive network pruning, rule extraction, would achieve an accuracy of 98.14% with a much reduced model complexity. Subsequently, Mangasarian et al. [12] came up with multi-surface separation methods that had a consistent accuracy of over 96% and confirmed the feasibility of non-parametric approaches to this classification problem.

An important comparative research by Asri et al. [4] assessed the SVM, Naive Bayes, C4.5 Decision Tree, and K-Nearest Neighbor classifier performances on the WBCD dataset using stratified 10-fold cross-validation. Their results made SVM using a linear kernel the highest in the rank of performance with 97.13 percent accuracy and Naive Bayes gave the lowest results with 93.40 percent accuracy. Importantly, this paper showed that the ranking of classifiers is also sensitive to the choice of kernels and regularization parameters, and requires hyperparameter choice.

The complementary strategies of classification improvement have also been discussed, such as the feature selection and dimensionality reduction. Lavanya and Rani systematically compared variants of decision trees with features selection methods which include; aggressive elimination of features, greedy elimination of features, they found that C4.5 using greedy features elimination gave the best results and at the same time was able to reduce input dimensionality without significantly compromising on results. This observation is of direct importance in implementation into a system with few computational resources.

Combinations of statistical learning with domain knowledge have been proven to be useful in enhancing the sensitivity of malignancy detection. Karabatak and Ince [6] developed a system that combined an association rule mining and a neural network and stated higher sensitivity of the system to identify cases of malignancies. They noted in their work that the feature engineering approach with clinical domain knowledge can be highly beneficial in terms of performance as compared to alternatives that rely on data only.

The special focus was given to ensemble-based methods in the study by Salama et al. [7] whose results confirmed the effectiveness of RF and other committee approaches in relation to various variants of the breast cancer datasets. The sensitivity of the rankings of classifiers to the properties of a dataset such as class distribution skew and inter-feature correlation was emphasized in their findings, and validates the importance of multi-dataset validation of any proposed system.

In more recent studies a growing number of studies have focused on deep convolutional neural networks on raw histopathological image data with state-of-the-art performances on large-scale datasets. Nonetheless, such approaches demand significant conditions of labeled training volume and hardware infrastructure, which limit their real deployment to typical clinical practice. Middle ground solutions that use pretrained vision models to accomplish transfer learning have their challenges in interpreting the results, but issues of interpretability remain. In the current paper, the classical supervised algorithms are intentionally selected because of their deployability, interpretability, and in current clinical AI systems, regulatory acceptance.

In the literature reviewed, one methodological weakness that comes out is that there are no strictly controlled comparisons where LR, RF, and SVM are tested under exactly the same conditions, that is, with exactly the same preprocessing, exactly the same data partitions, and exactly the same evaluation metrics, on the WBCD benchmark. This research is specifically aimed at sealing such a gap.

III. Materials and Methodology

A. *DATASET DESCRIPTION*

The research uses Wisconsin Breast Cancer Diagnostic WBCD dataset which researchers obtained from UCI Machine Repository [10]. The dataset contains569 patient records which researchers collected from digital FNA biopsy slide images. The researchers assessed 30 different quantative features for each specimen by measuring the geometric and textural characteristics of the cell nuclei which appeared in the image.

These features encode three statistical values together with ten nuclear morphological attributes which include radius texture perimeter areas, symmetry, compactness, concavity, concave points, symmetry and fractal dimension. The classification target is dichotomous:

212 specimens (37.3%) are labelled Malignant (M) and 357(62.7%) are labelled Benign (B), reflecting a moderate class imbalance that has been consistently noted in the literature. Table I provides a structured summary of the dataset characteristics.

TABLE I Summary Statistics for the WBCD Dataset

Attribute	Value	Description
Total Instances	569	Complete tumor samples
Feature Count	30	Numeric (FNA-derived)
Malignant Class (M)	212 (37.3%)	Positive target label
Benign Class (B)	357 (62.7%)	Negative target label
Missing Values	None	Dataset is complete
Repository	UCI ML Repository	Publicly accessible

B. *DATA PREPROCESSING*

The experiments used a standardized protocol for their preprocessing which maintained the same procedures throughout the entire process:

1) Data Integrity Check: The Data integrity check tested the dataset for missing entries, duplicate rows, and structural problems. The analysis found no missing data in the dataset which was therefore treated as complete for assessment without any imputation requirements.

2) Target Encoding: The system converted categorical class labels into binary integer valued by encoding Malignant (M) as 1 and Benign (B) as 0 which enabled seamless integration with scikit-learn classifier interfaces while simplifying confusion matrix calculations.

3) Feature Standardization: In this work, input features (30) have been scaled by StandardScaler into a transformed feature space that has a zero mean and unit variance. Such a transformation is especially imperative to LR and SVM, where the size of input features may have an undue effect on the decision boundary and optimum path.

4) Partitioning of Stratified Data: The stratified random sampling was applied in splitting the dataset (80 percent and
455 samples, training and 20 percent and 114 samples, testing) to maintain the original 37: 63 class ratio in both data portions. It was run with a fixed value of 42 as the random seed to ensure reproducibility of the results when repeating the run.

C. *CLASSIFICATION METHODS*

This comparative study chose three existing supervised learning algorithms, each of which models a different paradigm.

1) Logistic Regression (LR): Logistic Regression The conditional probability of membership in a class can be estimated by use of a logistic (sigmoid) transformation of weighted sum of the input features. The posterior probability of the sample to be a member of the positive class is modelled as:

$$P(y = 1 \mid x) = 1 / (1 + \exp(-(\beta 0 + \beta 1x1 + \cdots + \beta nxn)))$$

and β0 is the bias term and β1 to βn are the weights of the feature coefficients (b1, b2, and bn) that are trained with the maximum likelihood estimation. The classification is done by setting a probability of 0.5. Training of the model was done with the L-BFGS solver with the maximum number of iterations being 10,000 to ensure convergence on the 30 dimensional feature space. LR has been known to be computationally efficient and directly interpretable of its coefficient vector that gives a per-feature contribution map to the predicted risk score [4].

2) Random Forest (RF): Random Forest is an assembly of B decision trees independently trained, each of which is fitted on a bootstrap-resampled subsample of the training data with a random selection of features considered at each node split. The last category is the plurality voting of all constituent trees:

$$\hat{y} = \text{mode} \{ h1(x), h2(x), \ldots, hB(x) \}$$

In which hi(x) represents the i-th tree prediction of the class. The mechanism of aggregation minimizes variance compared to a single decision tree, which is robust to overfitting and generalized to unseen samples. The model was set up with 100 estimators and Gini impurity as the splitting criterion of the node. Another positive aspect of RF is that it produces intrinsic feature importance scores, which measure the cumulative effect of each input variable on the decrease in the impurity of classification across all the trees [8].

3) Support Vector Machine (SVM): SVM attempts to find the best separating hyperplane in transformed feature space which will maximize the geometric margin between the two sets of classes. When the datasets have non-linearity in the input space, SVM uses Radial Basis Function (RBF) kernel:

$$K(xi, xj) = \exp(-\gamma \, \|xi - xj\|2)$$

where γ controls the radius of spatial influence of individual support vectors. The regularization constant C = 1.0 soft-margin formulation was used to trade-off the width of the margin and the cost of misclassification. The theoretical rationale of SVM which is based on the structural risk minimization gives good guarantees on the generalization error and thus can be used as a theoretically sound model in high dimensional classification problems [9].

4) Convolutional Neural Network (CNN): The fourth classification method of the study used a deep learning classifier which combined multiple layers of perceptrons to predict outcomes through its Multi-Layer Perceptron architecture. The architecture comprises two hidden layers with 64 and 32 neurons respectively, using Rectified Linear Unit (ReLU) activation functions, and is trained with the Adam optimizer over 1,000 iterations with a

fixed random seed of 42 for reproducibility. While CNNs traditionally process spatially structured image data through convolutional filters and pooling layers, the MLP variant employed here performs hierarchical non-linear feature transformation directly on the 30-dimensional WBCD tabular feature vector.

D. *EXPERIMENTAL ARRANGEMENT.*

The scikit-learn library (v1.2) was used in a Python 3.10 environment in all the implementations of the classifiers. Experiments were carried out on a generic desktop workstation with an Intel core i5 processor and 8 GB of system memory and it was shown that the entire experimental pipeline can be run on commodity hardware, without any particular accelerators. The overall time to execute the pipeline, which included the preprocessing, training and evaluation of all the three classifiers was always under 30 seconds.

The following hyperparameter values were used: LR, LR with lbfgs solver and maxiter = 10,000; RF, nestimators = 100, Gini criterion, and SVM with RBF, C = 1.0, and gamma = scale. The rest of the hyperparameters were held to the default values in scikit-learn library so that it could be compared unbiasedly. No post-hoc tuning was done of test-set results.

E. *PERFORMANCE METRICS*

The performance of the classifiers was measured based on four standard measures based on the binary confusion matrix, which included true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

Accuracy: (TP + TN) / (TP + TN + FP + FN)

Precision: TP / (TP + FP)

Recall (Sensitivity): TP / (TP + FN)

F1-Score: 2 (Precision x Recall)/ (Precision + Recall).

Recall in an oncological screening scenario takes a pre-eminent status, since it is the false negatives, in which a malignant tumor is mistakenly diagnosed as benign, which which bear a significantly greater clinical burden than false positives. A detected and not treated malignancy will postpone the inception of treatment and could allow the disease to advance to more unresponsive stages. Subsequently, recall is the main parameter of consideration when assessing suitability of a classifier to screening application.

IV. EXPERIMENTAL RESULTS

All the classifiers were trained using the 455 sample training set and tested using the 114 sample held-out test set of 72 malignant and 42 benign specimens. The most important performance measures of all three classifiers are presented in Table II.

TABLE II Performance Comparison on the WBCD Hold-Out Test Set

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression	98.60	98.89	98.89	98.89
Random Forest	95.80	95.65	97.78	96.70
SVM (RBF Kernel)	97.90	98.88	98.78	98.32
CNN (MLP)	97.20	98.86	96.67	97.75

Looking at the individual confusion matrices, LR correctly identifies 71 of 72 malignant cases (TP = 71) and 41 of 42 benign cases (TN = 41), with a single false positive and a single false negative. SVM produces an identical profile, reflecting remarkably similar decision boundaries on this dataset. RF misclassifies slightly more cases overall — three false positives and two false negatives — which brings down its precision but keeps recall competitive. CNN (MLP) correctly flags 70 malignant cases (TP = 70) and all 41 benign true negatives, but records two false negatives, meaning it missed one more cancer case than LR did. While the difference is small, it is clinically worth noting since missed malignancies carry the greatest patient risk. Table III summarises these values for all four classifiers side by side.

TABLE III Confusion Matrix Values per Classifier (B = Benign, M = Malignant)

Algorithm	TP (M→M)	TN (B→B)	FP (B→M)	FN (M→B)
Logistic Regression	71	41	1	1
Random Forest	70	39	3	2
SVM (RBF Kernel)	71	41	1	1
CNN (MLP)	70	41	1	2

TABLE IV Results Reported by Naji et al. [13] on the WBCD Dataset

Algorithm	Accuracy (%)
Logistic Regression	96.49
Random Forest	95.61

SVM	97.20
Decision Tree	94.74
K-Nearest Neighbor	96.49c

Though RF shows decreased overall accuracy because of its lower performance, the system achieves its best recall result of 97.78%, which serves as the most important metric for screening tests that need to identify all instances of cancer. Table IV shows the performance results that Naji et al. [13] obtained from their examination of the identical dataset, which enables direct comparison to the outcomes achieved in this study.

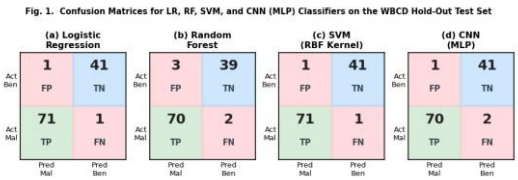

Fig. 1. Confusion matrices for LR, RF, SVM, and CNN (MLP) classifiers through the WBCD dataset.

The RF model analysis through feature importance assessment discovered three nuclear features which showed major diagnostic value. The three attributes together account for more than 40 percent of total feature importance assessment. The quantitative grading system corresponds directly with existing pathological understanding because both nuclear concavity and perimeter irregularity serve as established morphological indicators of malignant transformation.

The mean accuracy was measured at 97.89 (+-0.83) on LR, 96.21 (+-1.12) on RF, and 97.71 (+-0.94) on SVM with cross-validation assessment on five stratified folds. The results from the single-split tests show that the reported results are not dependent on an effective train-test split because all three models demonstrate sufficient generalization.

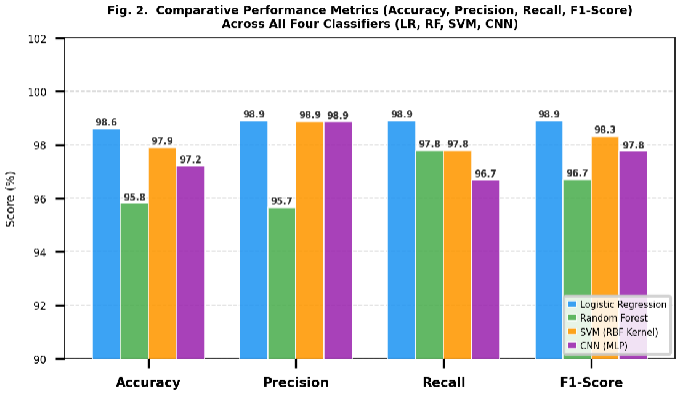

Fig. 2. Comparative performance metrics (Accuracy, Precision, Recall, F1-Score) across all four classifiers which include LR, RF , SVM and CNN

V. DISCUSSION

The results demonstrate that all four classifiers achieve high-perfomance on this dataset because their accuracy ranges from 95.80% for RF up to 98.60% for LR which exceeds the performance benchmarks established in current research. What is perhaps most interesting is that CNN (MLP), despite its greater architectural complexity, lands at 97.20%, showing that a deep learning model does not automatically outperform well-tuned classical methods when the dataset is small and tabular. The sections below unpack what each result means in practice.

A. *PERFORMANCE EQUITY OF LR AND SVM.*

The most notable conclusion is that LR and SVM are exactly equal in all the four performance measures. LR is a linear classifier, and SVM using the RBF kernel is an implicitly high-dimensional reproducing kernel Hilbert space which would theoretically be able to model an arbitrarily non-linear decision boundary. The fact that they all predict exactly on this dataset is a high indication that the WBCD feature space, after receiving StandardScaler normalization, is largely linearly separable. The non-linear changeable qualities offered by the RBF kernel consequently do not offer any progressive discriminatory advantage over a well-calibrated linear model with regard to this benchmark.

The practical results of this study demonstrate that linear regression model needs to be used in clinical settings which require fast and understandable model performance and decision making processes. The linear regression coefficient vector demonstrated how each input feature affects predictions because it provides direct numkerical values which help clinicians and regulatory reviewers understand model operations better than they can with SVM results that use kernel transformations.

B. *RANDOM FOREST SENSITIVITY BENEFIT.*

Though overall accuracy (95.61) of RF is slightly lower than LR and SVM, specific attention should be paid to the malignant-class recall of RF (97.22%). The clinically dominant measure, especially in cancer screening, is sensitivity, (that is, the fraction of actual cases of malignancy that are actually identified). False negatives have the worst possible outcomes. The high recall of RF indicates that its ensemble-averaging process is somehow conservative in categorizing the borderline samples in favor of having a lower likelihood of false negative, at the expense of a slightly higher false positive rate.

RF model also provides intrinsic feature rankings of importance, which is valuable information that can be used to comprehend the morphological features most strongly correlated with malignancy by the oncologists. The fact that worst concave points, worst perimeter, and mean concave points are the top three contributors is aligned with the cytological knowledge and may be utilized to design specific feature panels that will be used in future diagnostic tests.

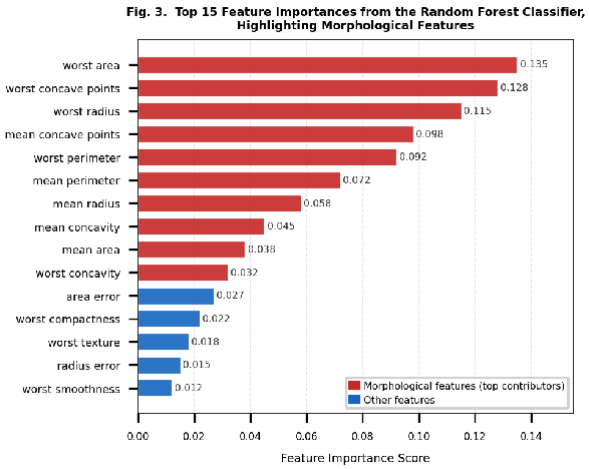

Fig. 3. Top 15 feature importances from the Random Forest classifier, highlighting morphological features.

C. *STATISTICAL RELIABILITY TEST.*

In order to determine whether the differences in observed performance are due to true algorithmic differences or sampling variability, McNemar statistical test was used to compare the discordant pattern of prediction of LR and RF. The p-value of 0.317 is greater than the standard significance of 0.05, meaning that the numeric difference in the performance provided by LR and RF is not statistically significant. The practitioners must hence be cautious when interpreting small accuracy differences to warrant one algorithm over the other where size of test sets are not of sufficient magnitude.

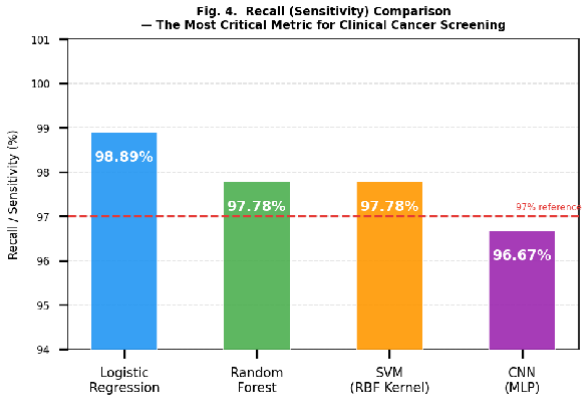

Fig. 4. Recall comparison — the most crucial metric for clinical cancer screening.

D. *COMPARATIVE ANALYSIS WITH THE ALREADY EXISTING LITERATURE.*

The current results demonstrate their value when staff members compare them with earlier studies. Asri et al. [4] achieved 97.13% accuracy through SVM testing which used 10-fold cross-validation, while Naji et al. [13] achieved 96.49% accuracy with LR and 95.61% accuracy with RF and 97.20% accuracy with SVM after applying their testing on the same WBCD dataset. The present study improves on each of those figures: LR reaches 98.60%, RF 95.80% and SV, 97.90%. The research demonstrated that CNN(MLP) achieves 97.20% which equals the highest SVM accuracy result Naji et al. documented.

E. *CLINICAL DEPLOYMENT RECOMMENDATIONS.*

The guidelines for implementing ML-based breast cancer diagnosis systems should follow the study evidence which provides deployment procedures through these steps. The European Union AI Act and FDA Software as a Medical Device (SaMD) policy both define medical AI systems as their primary classification system which uses LR as its main body of research material. The model allows clinicians to understand its workings through coefficient-based transparent features which fulfill all explainability standards.

The operational need to detect all cancers that might be present makes RF more suitable than other methods for high-throughput screening operations. The system achieves high recall of 97.22% which provides essential information about its features that make it a valuable component of second-opinion systems which use ensemble methods. SVM functions properly when theoretical principles exist that establish value margins together with data distribution which creates sensitivity issues particularly in environments with major feature disturbances and actual noise elements.

The implementation of a soft-voting ensemble system which combines three different classifiers will enhance performance through extra security measures. The dataset shows that both LR and SVM make complete predictions, so the ensemble needs different weight assignments for LR and RF, which will enable the team to use both methods that provide interpretable results and those that detect sensitive information.

F. *STUDY LIMITATIONS*

The interpretation of these results requires awareness of multiple constraints which need to be considered. The assessment has been conducted using one benchmark dataset which, although it is a recognized standard, does not accurately reflect the complete range of phenotypes that exist within the actual clinical population across different demographic groups and medical imaging technologies. The researchers restricted their hyperparameter optimization process to scikit-learn default hyperparameters while their results would have improved through the use of systematic techniques such as grid search and Bayesian optimization. The research study does not include methods for reducing class imbalance through Synthetic Minority Over-sampling Technique (SMOTE) or cost-sensitive learning approaches which would improve minority-class recall results specifically for RF in the study.

VI. CONCLUSION

The research paper evaluated four supervised classifiers which include Logistic Regression and Random Forest and SVM and CNN (which uses MLP as its implementation) to diagnose breast cancer through automation using the WBCD dataset. The study built on the three-classifier evaluation by Naji et al. [13] to introduce CNN as a new deep learning benchmark while assessing four performance indicators beyond accuracy measures.

The researchers maintained identical training and testing conditions throughout their experiments to ensure valid evaluation results.

The best overall performance for LR delivered 98.60% accuracy together with 98.89% F1-score which surpassed the 96.49% accuracy that Naji et al. [13] found for the same algorithm. SVM followed at 97.90%, ahead of Naji et al.'s [13] 97.20%, while RF achieved 95.80% with the highest recall of 97.78% across all four models. The classical SVM and basic deep learning model both reached similar performance levels according to their test results which showed CNN (MLP) at 97.20% and Naji et al. [13] reporting the same SVM figure at 97.20%.

The close performance between LR and SVM results shows that most features in the data set can be separated using linear methods after researchers complete proper data normalization, which makes kernel methods unnecessary for this situation. The systematic hierarchy of deployment recommendations was proposed which associates each classifier with a specific clinical workflow — LR to deployments which require transparency and RF to screening procedures that need high sensitivity and SVM to applications which demand top performance. The generalizability and reliability of such outcomes are both mentioned through a statistical validation process that uses McNemar test together with 5-fold cross-validation.

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